

Experiment No 3 Hybrid Cloud

Setting Up Hybrid Connectivity Experiment with setting up a secure VPN connection between an on-premises data center and a public cloud provider such as AWS or Azure.

Configure a hybrid network topology using virtual private clouds (VPCs), subnets, and route tables.

Steps:

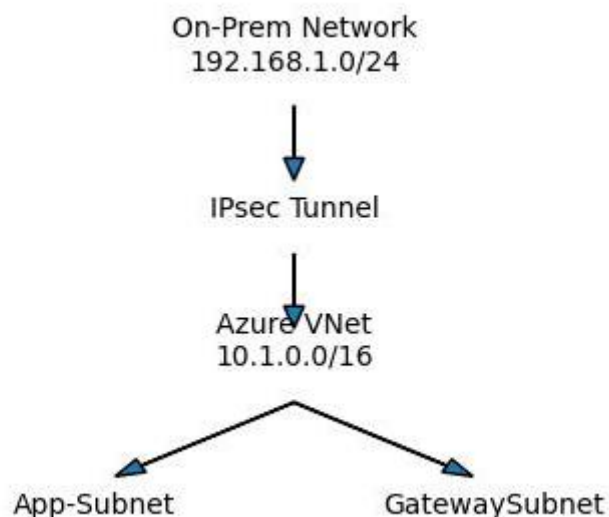
Create a Virtual Network (VNet)

Create Subnets

Create a VPN Gateway

Simulate an “On-Prem” network

Connect them using Site-to-Site VPN



STEP 1: Create a Resource Group

Think of Resource Group as a folder.

1. Login to Azure Portal
2. In search bar type **Resource groups**
3. Click → **Create**
4. Fill:

- Subscription → Your student subscription
- Resource group name → Hybrid-RG
- Region → Choose same region everywhere (example: Central India)

5. Click **Review + Create**

6. Click **Create**

STEP 2: Create Virtual Network (VNet)

VNet = Your cloud network.

1. Search **Virtual Networks**
2. Click **Create**
3. Fill:

Basics Tab

- Resource Group → Hybrid-RG
- Name → Cloud-VNet
- Region → SAME region as resource group

Click **Next**

IP Addresses Tab

- Address Space:

10.1.0.0/16

Remove default subnet.

Click + **Add Subnet**

Create Subnet 1:

- Name: App-Subnet
- Address range:

10.1.1.0/24

Click **Add**

IMPORTANT: Add Gateway Subnet

Click + **Add Subnet** again:

- Name: GatewaySubnet (must be EXACT name)
- Address range:

10.1.255.0/27

Click Add

Now click **Review + Create**

Click **Create**

VNet created

STEP 3: Create Virtual Network Gateway (VPN Gateway)

This is Azure side VPN device.

1. Search → **Virtual network gateways**
2. Click → Create

Fill details:

Basics:

- Resource Group → Hybrid-RG
- Name → Azure-VPN-Gateway
- Region → Same region
- Gateway Type → VPN
- ☐ VPN Type → Route-based
- ☐ SKU → Select VpnGw1 (basic supported)
- ☐ Generation → Generation1

Virtual Network Section

Select:

Cloud-VNet

Public IP Address

Click **Create new**

Name it:

VPN-Gateway-IP

Click OK

Click **Review + Create**

Click **Create**

⚠ IMPORTANT:

This takes **30–45 minutes** to deploy. Be patient.

STEP 4: Simulate On-Prem Network

Since you don't have real data center, we simulate.

We create:

Local Network Gateway (represents on-prem)

1. Search → **Local Network Gateways**
2. Click → Create

Fill:

- Resource Group → Hybrid-RG
- Name → OnPrem-Gateway
- Region → Same region

IP Address:

Put any dummy public IP (example):

1.1.1.1

Address Space:

Add:

192.168.1.0/24

Click **Review + Create**

Click **Create**

Done

Create VPN Connection

Now connect Azure to On-Prem.

1. Go to **Virtual Network Gateways**
2. Click your gateway → Azure-VPN-Gateway
3. On left side click **Connections**
4. Click + **Add**

Fill:

- Connection Name → Hybrid-Connection
- Connection Type → Site-to-Site (IPsec)
- Virtual Network Gateway → Auto selected
- ☐ Local Network Gateway → OnPrem-Gateway
- ☐ Shared Key → Type simple password:

Azure123

Click **OK**

VPN connection created

STEP 6: Verify Connection

1. Open Azure-VPN-Gateway
2. Click **Connections**
3. Check Status

If configured with real on-prem router → it will show Connected.

Since we used dummy IP, it will show Not Connected.

